1.1 Sample spaces and events

Probability starts with a model of an experiment: anything whose result we cannot predict in advance, such as rolling dice, dealing cards or drawing a name from a hat. Each possible result is an outcome, and the set of all outcomes is the sample space, written Ω.

Choosing Ω well is half the work. For two dice it pays to imagine one red die and one blue die, so that an outcome is an ordered pair (i, j): red shows i, blue shows j. There are $6 \cdot 6 = 36$ such pairs. If we recorded only the unordered result, such as “a one and a six”, there would be 21 results, but they would not be equally likely, because a one and a six can happen in two ways and a double in only one.

An event is a subset of Ω: a collection of outcomes, described by a property such as “the sum is 7”. The event happens when the outcome of the experiment belongs to it.

Events combine like sets. The complement E' contains the outcomes not in E. The union $E \cup F$ contains the outcomes in at least one of E and F, and the intersection $E \cap F$ the outcomes in both. Two events are disjoint when their intersection is empty, so they can never happen together.

1.2 Equally likely outcomes and the axioms

When Ω is finite and there is no reason to prefer one outcome to another, as with fair dice or a well-shuffled deck, we give every outcome the same probability $1/|\Omega|$. The probability of an event is then the fraction of outcomes it contains: $P(E) = |E| / |\Omega|$. For two dice, six of the 36 pairs have sum 7, so $P(\text{sum is } 7) = 6/36 = 1/6$.

Not every experiment has equally likely outcomes, so probability in general is defined by three axioms. Probabilities are never negative; the whole sample space has probability 1; and the probability of a union of disjoint events is the sum of their probabilities. The counting formula satisfies all three.

Everything else follows from the axioms. Since E and E' are disjoint and together make up Ω, $P(E) + P(E') = 1$, which is the complement rule. Taking $E = \Omega$ gives $P(\varnothing) = 0$.

For events that overlap, adding $P(E)$ and $P(F)$ counts the outcomes in $E \cap F$ twice, so we subtract them once: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$. This is inclusion–exclusion for two events. With two dice, “at least one six” has 11 outcomes, “doubles” has 6, and they share only (6, 6), so the union has $11 + 6 - 1 = 16$ outcomes.

1.3 The multiplication rule and permutations

Equally likely outcomes turn probability into counting, and most counting rests on one principle. If a task is done in steps, and the number of ways to do each step does not depend on how the earlier steps were done, then the number of ways to do the whole task is the product of the numbers for the steps.

A four-digit lock code has 10 choices for each digit, so $10^4 = 10{,}000$ codes. Four rolls of a die give $6^4 = 1296$ sequences. Only the number of choices must stay fixed; which choices are available may change from step to step.

Putting n different objects in a row is a task in n steps, with n choices for the first place, $n - 1$ for the second and so on, so there are n! orderings. Choosing an ordered list of k objects from n, with no repeats, stops after k steps: $n \cdot (n - 1) \cdot \ldots \cdot (n - k + 1) = n! / (n - k)!$. We write this nPk. Gold, silver and bronze among eight runners can go in $8 \cdot 7 \cdot 6 = 336$ ways.

The rule also counts complements. The four rolls with no six number $5^4 = 625$, so the probability of at least one six is $1 - 625/1296 = 671/1296$, a little over one half.

1.4 Combinations, and a look ahead

Often order does not matter: a committee, a hand of cards, a set of lottery numbers. Each unordered selection of k objects can be listed in k! orders, so the ordered count is k! times the unordered one. Dividing, the number of ways to choose k of n objects is nCk = n! / (k! (n − k)!), also written C(n, k) and read “n choose k”.

A committee of three from ten people can be chosen in C(10, 3) = 120 ways, and a five-card hand from a 52-card deck in C(52, 5) = 2,598,960 ways. Choosing who is in a selection is the same as choosing who is left out, so C(n, k) = C(n, n − k).

Counting answers surprising questions. In a group of n people, ignoring leap days and assuming all 365 birthdays equally likely, the chance that all birthdays differ is 365 · 364 · … · (365 − n + 1) / 365^n, by the multiplication rule on top and bottom. The complement, at least one shared birthday, passes one half at n = 23.

The answer feels too large because we picture someone matching our own birthday. With 23 people there are C(23, 2) = 253 pairs, and any pair can match.

2.1 Conditional probability

Often we learn something before the whole outcome is known: the die landed on an even number, the first card was an ace, the test came back positive. Conditional probability says how the probabilities should change.

Definition. If $P(B) > 0$, the conditional probability of A given B is $P(A \mid B) = P(A \text{ and } B) / P(B)$. Knowing that B happened, we keep only the outcomes inside B and rescale their probabilities so that they add up to one again. For a fair die, $P(\text{six} \mid \text{even}) = (1/6) / (1/2) = 1/3$: the six is one of three equally likely even faces.

For a fixed B, $P(A \mid B)$ as a function of A is itself a probability. It is never negative, gives B probability one and adds over disjoint events. Every rule from Lecture 01 therefore holds conditionally, for example $P(A' \mid B) = 1 - P(A \mid B)$, where A' is the complement of A.

Multiplying the definition out gives the multiplication rule $P(A \text{ and } B) = P(B) \cdot P(A \mid B)$, and repeating it gives the chain rule $P(A \text{ and } B \text{ and } C) = P(A) \cdot P(B \mid A) \cdot P(C \mid A \text{ and } B)$. The chain rule suits draws made one after another: the chance that two cards drawn without replacement are both aces is $(4/52) \cdot (3/51) = 1/221$.

2.2 Independence

Two events are independent when $P(A \text{ and } B) = P(A) \cdot P(B)$. When $P(B) > 0$ this is the same as $P(A \mid B) = P(A)$: being told that B happened does not change the probability of A. The definition is symmetric in A and B and also makes sense when one of the probabilities is zero.

Independence is a numerical property, so it has to be checked, not guessed from the story. With two fair dice, the event that the first die shows 6 has probability 6/36, the event that the sum is 7 has probability 6/36, and both happen only for the outcome (6, 1), with probability $1/36 = (1/6) \cdot (1/6)$. The two events are independent, even though the first die clearly takes part in the sum. Replace the sum 7 by the sum 8 and the product rule fails: 1/36 is not $1/6 \cdot 5/36$.

A common mistake is to confuse independent with disjoint. If A and B cannot both happen and each has positive probability, then $P(A \text{ and } B) = 0$ while $P(A) \cdot P(B) > 0$, so they are not independent. In fact they are as dependent as events can be: learning that B happened tells us for certain that A did not.

If A and B are independent, so are A and B', A' and B, and A' and B'.

2.3 Total probability and Bayes' rule

A partition of the sample space is a collection of disjoint events B1, B2, … whose union is the whole space: exactly one of them happens. The law of total probability says P(A) = P(A | B1) P(B1) + P(A | B2) P(B2) + …, with one term for each piece. It follows from the multiplication rule, because the events "A and Bk" are disjoint and together make up A.

The simplest partition is B together with its complement B′, which gives P(A) = P(A | B) P(B) + P(A | B′) P(B′). It splits one hard question into two easier ones: what if B occurs, and what if not?

Bayes' rule reverses a conditional probability. Writing P(A and B) in two ways, as P(B | A) P(A) and as P(A | B) P(B), and dividing by P(A) > 0 gives P(B | A) = P(A | B) P(B) / P(A); the denominator usually comes from total probability. P(B) is called the prior, P(A | B) the likelihood and P(B | A) the posterior.

A tree diagram organises the same calculation. Its first level of branches is the partition, labelled with the prior probabilities; the second level is the observation, labelled with conditional probabilities. Multiplying along a path gives its probability; Bayes' rule divides one path by the sum of all paths ending in the same observation.

2.4 Base rates

A diagnostic test is described by three numbers: the prevalence P(D) of the condition, the sensitivity P(+ | D), and the false-positive rate P(+ | D′). A person who tests positive wants the positive predictive value P(D | +), which depends on all three.

Take a prevalence of 1%, a sensitivity of 99% and a false-positive rate of 5%. Bayes' rule gives P(D | +) = (0.99 · 0.01) / (0.99 · 0.01 + 0.05 · 0.99) = 0.0099 / 0.0594 = 1/6, about 16.7%. In natural frequencies: of 10,000 people, 100 have the condition and 99 of them test positive; of the 9,900 who do not, 495 test positive anyway. Only 99 of the 594 positives are sick.

Treating the sensitivity as if it were P(D | +), and so ignoring the base rate, is called the base-rate fallacy. When the condition is rare, the healthy group is so much larger that even a small false-positive rate produces more positives than the sick group does.

Evidence can be used again. After one positive result the posterior 1/6 becomes the prior for a second, independent test, and a second positive raises the probability to 99/124, about 79.8%.

3.1 Random variables and their distributions

Many questions about a random experiment are really questions about a number computed from its outcome: the total shown by two dice, the number of heads in a run of flips, the amount won in a game. A random variable makes this precise. It is a function X that assigns a real number $X(\omega)$ to each outcome ω of the sample space. The function itself is fixed; what is random is the outcome fed into it.

Every statement about X describes an event. The event $\{X = k\}$ is the set of outcomes that X sends to k, so its probability is already determined by the probability on outcomes. For two fair dice and the sum S, the event $\{S = 7\}$ contains the six outcomes $(1, 6), (2, 5), \ldots, (6, 1)$, so $P(S = 7) = 6/36 = 1/6$.

When X takes finitely or countably many values, its probability mass function, or PMF, lists $p(k) = P(X = k)$ for each value k. A PMF has no negative entries and its entries sum to 1. The PMF forgets which outcomes produced each value: the first die alone and a single die rolled on its own have the same PMF, as do the minimum of two dice and seven minus their maximum.

3.2 Expected value

The expected value of X is the average of its values, each weighted by its probability: $E[X] = \Sigma\, k \cdot P(X = k)$. Grouping the outcomes by the value of X shows that the same number is $\Sigma\, X(\omega) \cdot P(\{\omega\})$, summed over outcomes. For one fair die, $E[X] = 21/6 = 7/2$. The die never shows 3.5; the expected value is a long-run average, not a prediction of any single roll.

To average a function of X there is no need to work out the PMF of g(X) first: $E[g(X)] = \Sigma\, g(k) \cdot P(X = k)$. For a die, $E[X^2] = 91/6$. Note that this is not $(E[X])^2 = 49/4$. Averaging and then squaring is not the same as squaring and then averaging.

Expected values give a natural notion of fairness. If a bet costs 1 and pays 6 when a die shows a six, the net gain is 5 with probability 1/6 and -1 with probability 5/6. Its expected value is $5/6 - 5/6 = 0$, so the bet is fair. Fairness says nothing about risk: a player still loses most rounds, and the gains come in rare large amounts.

3.3 Joint distributions and linearity

Two random variables on the same experiment have a joint PMF, $p(x, y) = P(X = x \text{ and } Y = y)$. Summing a row or column recovers a marginal PMF: $P(X = x) = \Sigma_y\, p(x, y)$. The marginals alone do not fix the joint PMF. X and Y are independent when every cell factors as $P(X = x) \cdot P(Y = y)$. Two dice rolled separately are independent; the first die X and the maximum M of both are not, since $X = 6$ forces $M = 6$.

Expectation is linear: $E[X + Y] = E[X] + E[Y]$ and $E[aX + b] = a \cdot E[X] + b$. The proof is one line, splitting the sum over outcomes, and it uses no independence at all. For the sum of two dice this gives $E[S] = 7$ without computing the PMF of S.

Linearity is strongest when combined with indicators. The indicator of an event A equals 1 when A occurs and 0 otherwise, so its expected value is $P(A)$. Shuffle n items and count those that stay in their original place. Position i keeps its item with probability $1/n$, so the expected count is $n \cdot 1/n = 1$, for every n, even though the indicators depend on one another.

3.4 Variance and spread

Two random variables can share an expected value and still behave very differently, so we also measure spread. With $\mu = E[X]$, the variance is $\mathrm{Var}(X) = E[(X - \mu)^2]$, the average squared distance from the mean. Expanding the square gives the shortcut $\mathrm{Var}(X) = E[X^2] - \mu^2$. For one die, $\mathrm{Var}(X) = 91/6 - 49/4 = 35/12$. The standard deviation $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$ is in the units of X; for a die it is about 1.71.

Shifting a random variable moves its mean but not its spread, and scaling by a multiplies the variance by a^2: $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$.

For sums, means always add but variances add only under extra conditions. If X and Y are independent, $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$, so the sum of two dice has variance 35/6. Without independence the rule can fail in either direction: $\mathrm{Var}(X + X) = 4\,\mathrm{Var}(X)$, while $X + (7 - X)$ is always 7 and has variance 0. The fair bet of 3.2 has mean 0 but variance 5.

4.1 Bernoulli trials and the binomial count

Many experiments repeat one simple trial: flip a coin, take a shot, test a part. A Bernoulli trial has only two outcomes, which we call success and failure, with probabilities p and 1 − p. We speak of repeated trials when the trials are independent and every trial has the same success probability p.

Each trial comes with an indicator I that equals 1 on a success and 0 on a failure. Its mean is E[I] = 1·p + 0·(1 − p) = p, and since $I^2 = I$ its variance is $p - p^2 = p(1 - p)$. It is largest, 1/4, when p = 1/2.

Now let X count the successes in n repeated trials. A particular sequence with k successes and n − k failures has probability p^k (1 − p)^(n − k), because the trials are independent. Every such sequence has the same probability, and there are C(n, k) of them, one for each choice of which k positions hold the successes. Adding them up gives the binomial distribution: P(X = k) = C(n, k) p^k (1 − p)^(n − k) for k = 0, 1, …, n. We write X ~ Bin(n, p).

The binomial theorem shows that these probabilities add up to (p + (1 − p))^n = 1. The case n = 1 is a single Bernoulli trial.

4.2 Mean, variance and waiting times

The mean of a binomial count has a short derivation. Write X = I_1 + ⋯ + I_n with one indicator per trial. Linearity of expectation, which needs no independence, gives E[X] = np. Because the trials are independent the variances also add, so Var(X) = np(1 − p). The standard deviation √(np(1 − p)) grows only like the square root of n.

Examples. Ten flips of a fair coin give exactly five heads with probability 252/1024, about 0.246: the most likely count, yet less likely than not. A player who scores 80% of free throws makes all of five shots with probability $0.8^5 \approx 0.328$. If each of 20 parts is defective with probability 0.05, independently, none is defective with probability $0.95^{20} \approx 0.358$.

A different question is how long we wait for the first success. Let T be the number of trials up to and including it. The first success comes at trial k exactly when k − 1 failures are followed by a success, so P(T = k) = (1 − p)^(k − 1) p for k = 1, 2, … This is the geometric distribution, and its mean is 1/p. Rolling for a six takes six rolls on average. The geometric distribution is memoryless: having failed many times does not shorten the wait still ahead.

4.3 Running averages and the law of large numbers

Watch a long run of trials and keep track of the proportion of successes so far, X/n. This running proportion is an average of indicators, so its mean is p and its variance is p(1 − p)/n. The spread of the proportion therefore shrinks like 1/□n: four times as many trials halve it.

The law of large numbers turns this into a statement about probabilities. For every tolerance ε > 0, the chance that the proportion is off from p by ε or more goes to zero: P(|X/n − p| □ ε) □ 0 as n grows. In words, the running proportion settles near p. This is what makes the long-run frequency interpretation of probability consistent with the rules we have been using.

The law says nothing of the kind about the count itself. The difference X − np between the number of successes and its expected value typically grows like □n. After 100 fair flips it is usually within about 5 heads of 50; after a million flips a gap of several hundred heads is ordinary. The gap grows, but more slowly than n, so divided by n it disappears.

A plot of the running proportion shows this clearly: wild swings in the first few dozen trials, then a curve that stays inside an ever narrower band around p.

4.4 Streaks and the gambler's misconception

After a fair coin lands heads five times in a row, many people feel that tails is now due. It is not. Independent trials have no memory, so the chance of heads on the next flip is still 1/2, whatever happened before.

How, then, does the proportion of heads come back to 1/2? Not by correction but by dilution. Suppose the first 10 flips were all heads. From then on we expect half of the remaining flips to be heads, so after n flips the expected number of heads is $10 + (n - 10)/2$. The expected excess over $n/2$ stays at 5 heads forever: nothing pays it back. Yet the expected proportion is $1/2 + 5/n$, which is 0.55 after 100 flips, 0.505 after 1000 and 0.5005 after 10 000. The streak is swamped by the flips that follow, not cancelled by them.

The same reasoning applies to roulette wheels, lottery numbers that have not come up in a while and a shooter who has missed several in a row: if the trials are truly independent with a fixed p, past results carry no information about the next one. When people do see real hot or cold streaks, the right conclusion is that the trials may not be independent, or that p may be changing, not that the future must balance the past.